HW4AI · ECE 410/510

CF01 CMAN

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(a) Network 784 → 256 → 128 → 10

| Layer | Input | Output | MAC |
|---|---|---|---|
| Layer1 | 784 | 256 | 784 × 256 = 200704 |
| Layer2 | 256 | 128 | 256 × 128 = 32,768 |
| Layer3 | 128 | 10 | 128 × 10 = 1280 |
| | | | |

(b)

Total MACs = MACs(layer1) + MACs(layer2) + MACs(layer3)

= 200704 + 32768 + 1280

= 234,752

(c) params = input_size × output_size

Total params = 784 × 256 + 256 × 128 + 128 × 10

= 234,752

(d) Weight memory = total params × 4 bytes

= 234752 × 4

= 939,008 bytes

(e) Activation memory = (784 + 256 + 128 + 10) × 4

= 4712 bytes

1f)

$$\text{Arithmetic Intensity} = \frac{2 \times \text{total MACs}}{\text{weight bytes} + \text{Activation bytes}}$$

$$= \frac{2 \times 234752}{939008 + 4712}$$

$$= \frac{469504}{943720}$$

$$= 0.497 \text{ flops/byte}$$